

Design Pack

Ingestion and scoring pipeline, class diagram, route deviation activity diagram

Project	Course	Case Study	Version
HaulTrack - 1,800-truck fleet telematics	B.Tech CSE - Software Engineering & Project Management	No. 19	1.0 (baseline)

1. Design goals

- **Fairness by construction:** terrain normalisation is a pipeline stage and a versioned parameter set, not a tweak inside the score formula.
- **Privacy by design:** location is used to compute events and deviations, then aged out; the insurer only ever sees aggregates.
- **Explainable:** every score can be decomposed to events with their gradient band and threshold, so a driver can see why.

2. Telemetry sizing

Assumptions are stated; the fleet and reporting interval are the given project data. Worst case assumes all trucks report 24 hours a day (real ignition-on time will be lower, giving headroom).

Quantity	Calculation	Result
Reports per truck per day	86,400 s / 30 s	2,880
Fleet records per day	1,800 x 2,880	5,184,000
Average ingest rate	5,184,000 / 86,400	60 records/s (about 3,600 per minute)
Design peak (assumed 3x average, shift-start sync)	3 x 60	180 records/s
Record size (assumption)	about 200 B stored incl. index overhead	200 B
Daily raw volume	5,184,000 x 200 B	1.04 GB/day (about 0.21 GB/day after roughly 5:1 compression)
90-day raw retention	1.04 GB x 90	93 GB uncompressed (about 19 GB compressed)
Event store (13 months, assumption: 400 km/day/truck, 3 harsh events per 100 km, 1 KB each)	1,800 x 400 km x 3% x 395 d x 1 KB	about 8.5 GB
Aggregates (36 months)	daily rows per depot / terrain class / metric	under 1 GB

Ingest at 60-180 records/s is modest for a partitioned queue, so the main design effort goes into correctness (map-matching, terrain lookup) and governance, not raw throughput. The retention periods are defined and justified in Deliverable 06.

3. Pipeline architecture

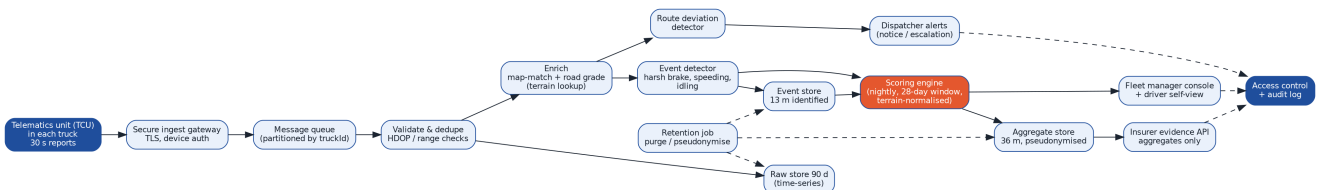


Figure 1 - Ingestion and scoring pipeline. Dashed lines are governance controls (retention, audit).

Stages. (1) The gateway authenticates the truck unit and accepts reports; (2) records are queued per truck; (3) validation drops duplicates and out-of-range or low-quality GPS fixes; (4) enrichment map-matches to the road and attaches gradient and speed limit; (5) the event detector flags harsh braking using the gradient-dependent threshold, and the deviation detector runs in parallel; (6) the nightly scoring engine computes a 28-day, terrain-normalised score

from events and exposure; (7) outputs go to the manager console, driver self-view and dispatcher; the insurer API serves aggregates only. The retention job and audit log wrap everything.

Scoring model (v1.0)

```
Score = 100 - (B + S + I)
B = min(50, 5 x h) h = terrain-adjusted harsh brakes per 100 km
S = min(35, 3.5 x q) q = % of driving time more than 10% over limit
I = min(15, 1.5 x z) z = % of engine-on time in excess idling
Rated only if distance >= 500 km in the 28-day window and unknown-gradient distance <= 5%.
Route deviation, fuel level and location are NOT inputs to the score.
```

A brake event counts when net deceleration $\geq$ the threshold of its downhill-gradient band (TN-2 in Deliverable 01). This is where the previous vendor failed: it applied one flat threshold everywhere.

4. Class diagram

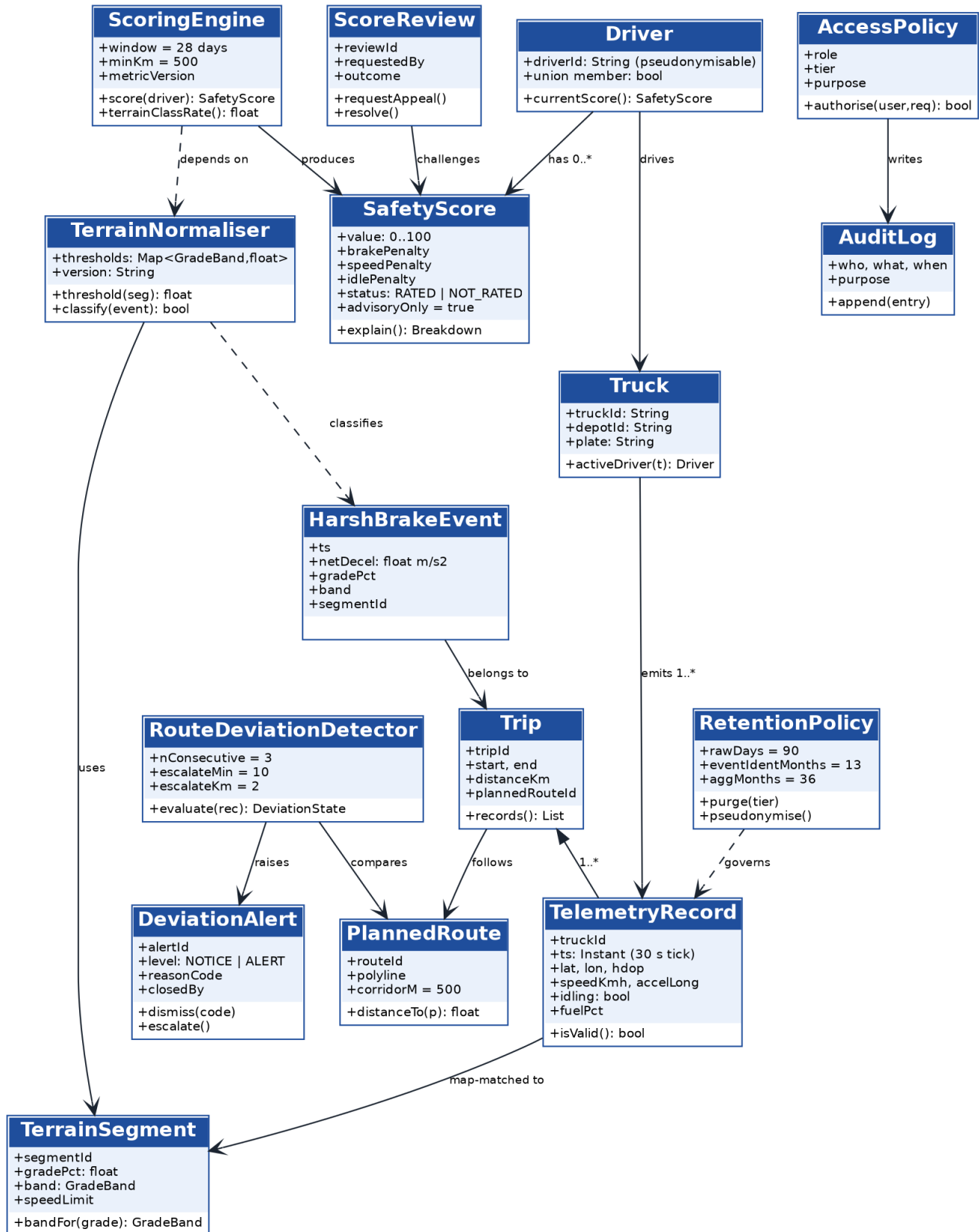


Figure 2 - Class diagram: telemetry, terrain normalisation, scoring, deviation detection and governance classes.

5. Activity diagram - route deviation detection

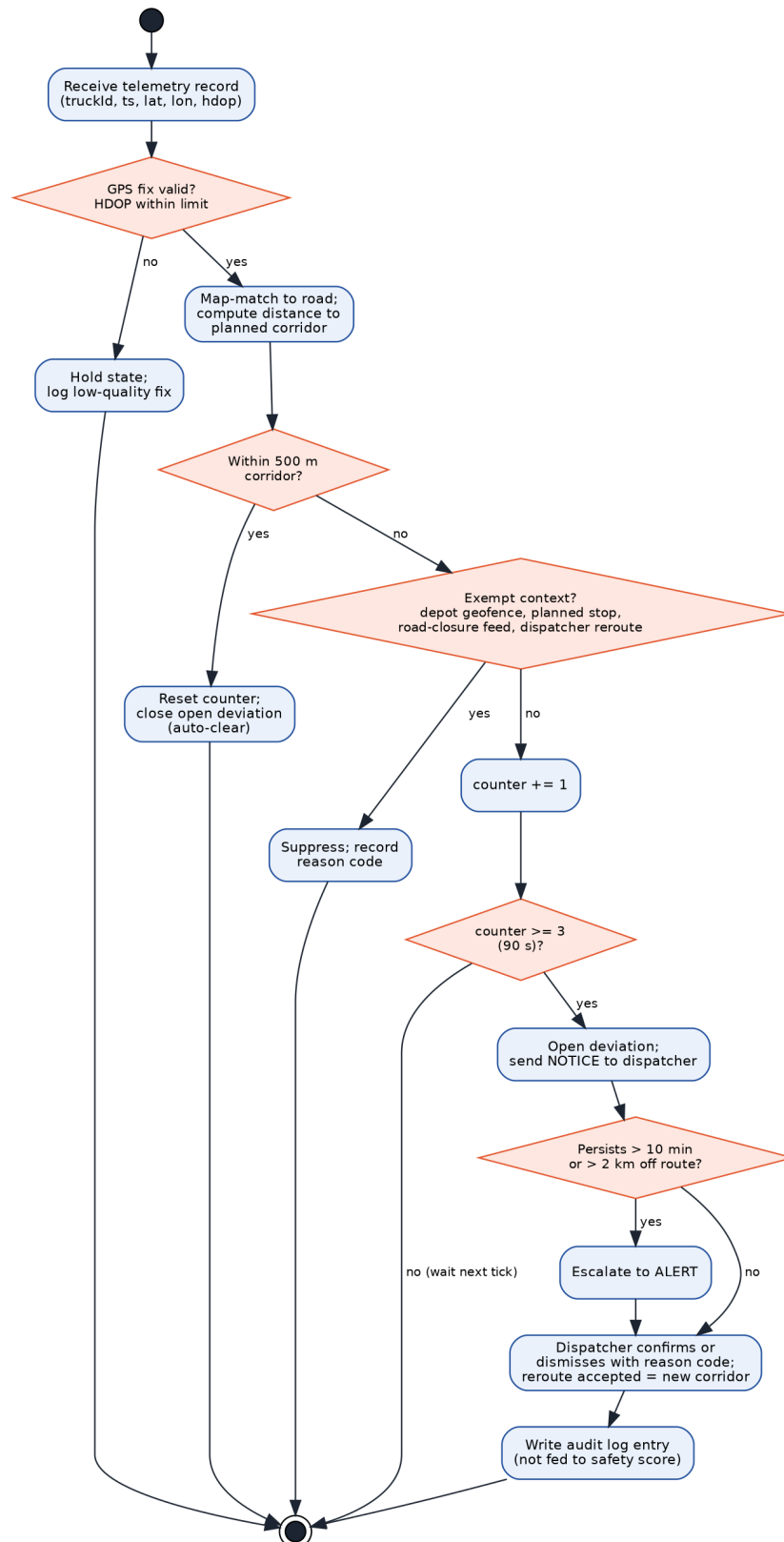


Figure 3 - Activity diagram for route deviation detection (defaults: 500 m corridor, 3 consecutive reports, escalate at 10 min or 2 km).

Design notes. Three consecutive 30-second reports (90 s) are required so one GPS jump cannot raise an alert. Exempt contexts (depot geofences, planned stops, road-closure feeds, dispatcher-approved reroutes) are checked before counting. The alert closes in an audit entry and is deliberately **not** an input to the safety score (decision D6).

Design decisions and justification

Decision	Reason
Terrain lookup is a separate, versioned stage	The vendor failure was a requirement gap; isolating terrain makes it testable and auditable.
Nightly batch scoring on a 28-day window	Scores change slowly; reduces noise and avoids reacting to one bad day.
Minimum 500 km exposure	Prevents unstable scores from short samples.
Pseudonymised aggregates for insurer	Meets I1/I2 without exposing named drivers (D2).
Deviation detector separate from scoring	Keeps operational alerts away from disciplinary use (D6).